

Milestone 2 – NBA Data Story

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Live website: com-480-data-visualization.github.io/airball

Project Goal

Our project, *Airball*, tells the story of how the NBA transformed over 78 years through four interactive data visualizations. We target a broad audience — from casual fans to data-curious analysts — and organize the narrative around four acts, each answering a distinct question about the evolution of basketball.

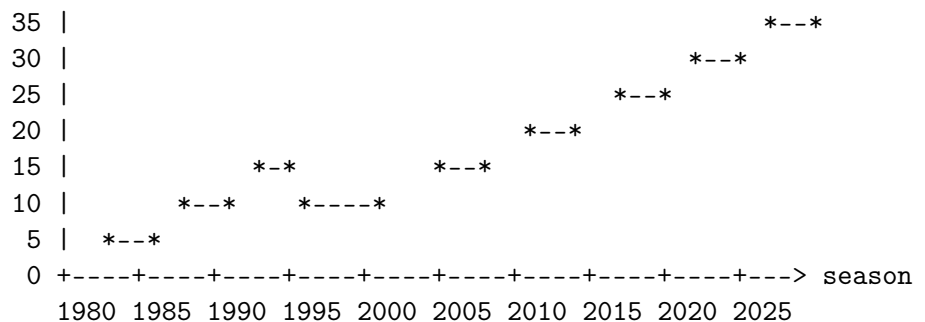
1. **Act I – The Revolution:** How did the three-point shot go from a gimmick to the dominant strategy in basketball?
2. **Act II – Era Explorer:** How did individual player efficiency and usage evolve across every season from 1982 to 2026?
3. **Act III – Player vs. Player:** How do the greatest players of different eras compare when normalized for era context?
4. **Act IV – Dynasties:** Which franchises dominated the NBA, and how do their win trajectories reveal the rise and fall of each dynasty?

In addition to the core visualizations, we plan to explore several more creative extensions that enhance interactivity of the project. These features are designed as optional additions and can be omitted without affecting the core narrative:

1. **Outlier Detection (Act II extension):** Highlight players who significantly outperform their era based on efficiency and usage metrics. This allows users to quickly identify exceptional or “ahead-of-their-time” performances.
2. **Dynasty Stability Metric (Act IV extension):** Introduce a custom metric combining win percentage, consistency, and longevity to better characterize and compare the strength and stability of different dynasties to visualize the difference between “flashy peak” vs “long-term dominance”
3. **“Build Your Player” Simulation (Act II/III extension):** Allow users to interactively define a hypothetical player (e.g., efficiency, usage, minutes) and visualize how such a profile would compare across different eras or against real players.

Visualization Sketches

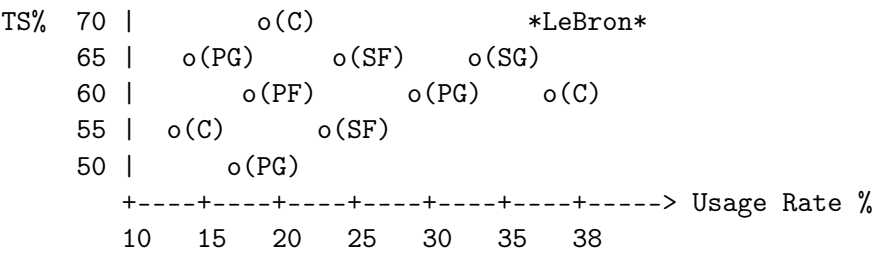
Act I — Annotated Line Chart (3PA over time)



[1979]	[1994-97]	[2012]	[2016]	[2024]
3pt line	Arc short.	Analytics	Warriors	New normal
adopted	(temp.)	revolution	dynasty	35+/game

Left panel: clickable narrative blocks per era. Right panel: sticky SVG line chart that animates the orange highlighted segment as the user clicks each block. Annotations appear progressively on the chart.

Act II — Animated Bubble Chart (Era Explorer)

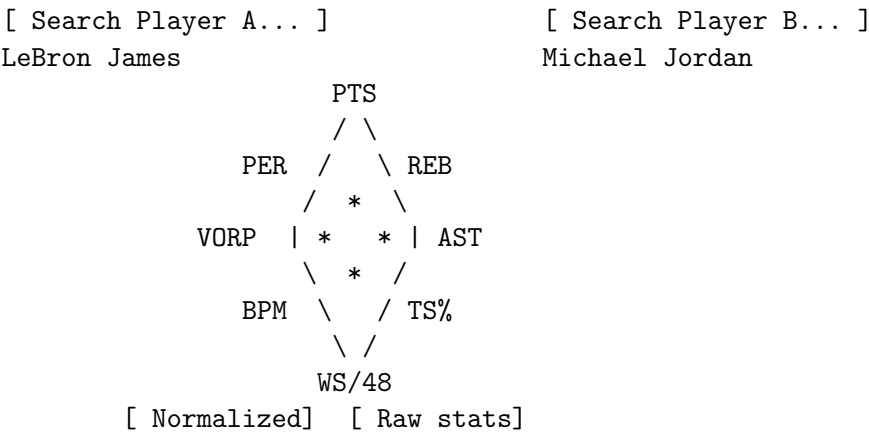


[Play] [slider: 1982 2026] Year: 2015

Bubble size = minutes/game Color = position

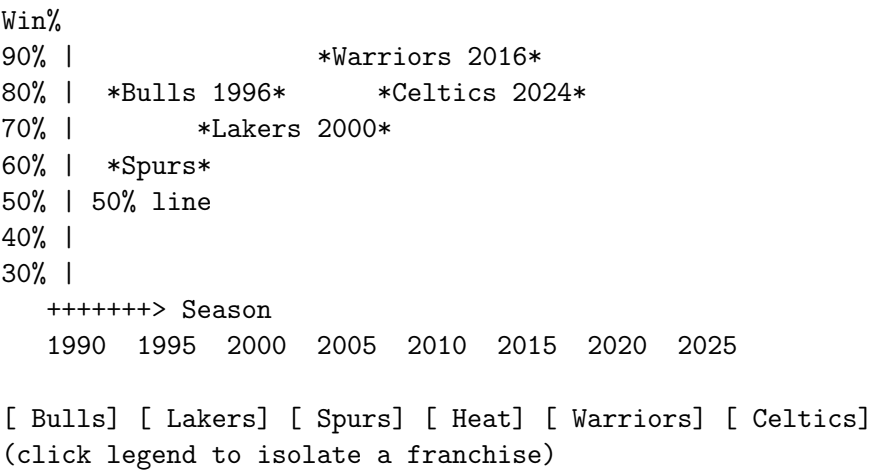
A year slider (with auto-play) filters to the 30 highest-minute players of that season. Hovering a bubble shows name, position, PPG, TS%, and USG%. A search box highlights a player across all years.

Act III — Radar Chart (Player Comparison)



Two searchable player inputs (2190+ players). Toggle between normalized percentile ranks (fair cross-era comparison) and raw career averages. Stat tables show winner highlighted in each player's color.

Act IV — Dynasty Line Chart



Six franchise lines (Bulls, Lakers, Spurs, Heat, Warriors, Celtics), each in a distinct color. Clicking a team in the legend highlights its line. Hovering shows season, team, and win percentage.

Tools and Lectures

Visualization	Tools / Libraries	Relevant Lectures
Act I – Line chart	Vanilla JS, SVG	Lec. 4 (Visualization design), Lec. 5 (Data & stories)
Act II – Bubble chart	Vanilla JS, SVG	Lec. 6 (Interaction), Lec. 7 (Animation)
Act III – Radar chart	Vanilla JS, SVG	Lec. 4 (Design), Lec. 8 (Perception & color)
Act IV – Line chart	Chart.js	Lec. 4, Lec. 7 (Animation)
Data pipeline	Python, NumPy, pandas,	Lec. 2 (Data), Lec. 3 (D3 / web stack)

Data is extracted from the [Kaggle NBA Stats dataset](#) (compiled from Basketball-Reference.com) using a single Python script (`scripts/extract_data.py`) that outputs four JSON files loaded by the website. No external visualization framework is used for Acts I–III; all SVG rendering is hand-coded in JavaScript for full control over animation and interactivity.

MVP and Extra Features

Core MVP

- **Act I:** Annotated line chart of league-average 3PA/game (1980–2026), with 5 clickable narrative blocks that progressively animate annotations on the chart.
- **Act II:** Animated bubble chart (Usage Rate vs. True Shooting%, bubble size = minutes/game, color = position), filterable by season via a slider with auto-play, and a player search highlight.
- **Act III:** Radar chart comparing two searchable players across 8 career metrics, with normalized (percentile) and raw modes, and highlighted stat tables showing which player wins each category.
- **Act IV:** Multi-line chart showing win% trajectories for 6 dynasty franchises (1990–2026), with interactive legend to isolate franchises.
- **Data pipeline:** A reproducible Python script that reads raw CSV files from Kaggle and generates the four JSON files consumed by the site.

Extra Features

- **Shot-zone heatmap (Act I extension):** Using `Player Shooting.csv`, add a court diagram showing the decline of mid-range attempts by zone over time. This requires the D3 lecture (Lec. 3) and geometric SVG work.
- **Draft predictor (Act V):** A heatmap of draft pick position vs. career VORP, revealing which draft slots reliably produce stars. Requires merging `Draft Pick History.csv` with advanced stats.
- **Salary vs. VORP (Act IV extension):** Adding a second tab to Act IV showing player salary vs. VORP when a salary dataset is integrated.
- **Scroll-driven narrative:** Replace the section-switching navigation with a smooth scroll-based storytelling flow using the Intersection Observer API, making the experience feel more like a data journalism article.